Ibrahim Kabbash

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Summary

DevOps Engineer with 3+ years of experience in Kubernetes, Linux, CI/CD, and systems reliability across Azure, GCP, and AWS. Held sole ownership of multiple concurrent client-facing, mission-critical production environments, with hands-on experience in AI workloads, security hardening, observability, and automation. Interested in platform engineering and driven by a constant need to go deeper and continuously improve.

Skills

- **Containerization & Orchestration**: Docker, Kubernetes, Helm, Cilium, Gateway API, Talos Linux
- **CI/CD Tools**: Azure DevOps, Jenkins, Argo CD, GitLab CI, CircleCI
- **Cloud Platforms**: Azure, AWS, GCP
- **IaC & Configuration Management**: Terraform, Ansible
- **Monitoring & Observability**: Prometheus, Grafana, Loki, Sentry, Splunk
- **Security**: Vault, Trivy, RBAC, Vulnerability Management, Container Security, Security Hardening, OAuth 2.0, SSO
- **Databases**: SQL Server, PostgreSQL, MySQL
- **Messaging & Streaming**: Kafka, Kafka Connect
- **Programming & Scripting**: Python, Bash

Certifications

- **Certified Kubernetes Security Specialist (CKS)**
- **Certified Kubernetes Administrator (CKA)**
- **GCP Certified Associate Cloud Engineer**
- **Microsoft Certified Azure AI Fundamentals**
- **AWS Cloud Practitioner**

Experience

- **eSpace** — Alexandria, Egypt
  Mid-Level DevOps Engineer — *02/2025 – Present*
  - Provisioned and managed multiple production GKE clusters with Terraform, integrating Cloud SQL, Filestore, and Identity-Aware Proxy for secure, cost-optimized multi-tenant deployments serving 5,000+ concurrent users.
  - Managed deployment and integration of 20+ services (Kafka, Kafka Connect, Elasticsearch, FastAPI, Spring Boot), collaborating with development teams and building Azure DevOps CI/CD pipelines to standardize delivery.
  - Collaborated with the development team on a UK-based AI cybersecurity project, providing SIEM knowledge-sharing and integration guidance to translate client requirements into implementation decisions.
  - Prototyped SIEM integrations (Azure Sentinel, Splunk) streaming security incidents into Kafka; designed and validated ingestion pipelines for reliable, fault-tolerant data delivery.
  - Designed Kafka and Kafka Connect scaling strategies for high-volume SIEM log ingestion, establishing a throughput-optimized pipeline for downstream services.
  - Stabilized GPU-enabled LLM infrastructure on Azure VMs and migrated from Ollama to vLLM, improving inference throughput and scalability for critical AI workloads.
  - Proposed and deployed a custom Qwen MoE model on AWS Bedrock over SageMaker, meeting internal model serving requirements at lower cost.
  - Onboarded an internal Angular and Spring Boot application onto EKS, creating Jenkins CI/CD pipelines and GitOps-based continuous deployment with Argo CD.
  - Supported the client development team in integrating Azure AI services including Azure OpenAI, Computer Vision, and AI Search, resolving integration blockers and providing technical guidance throughout the process.

DevOps Engineer *02/2023 – 01/2025*

- Owned full lifecycle of dedicated per-client Kubernetes environments (on-prem and AKS) for a .NET microservices platform, from cluster bootstrapping and application deployment to monitoring and production incident resolution.
- Built and owned Helm charts for a .NET microservices platform, migrating all client environments from raw Kubernetes manifests to standardized deployments, ensuring consistent releases across all environments.
- Led resolution of high-impact production incidents by coordinating root cause analysis across engineering and client-facing customer success teams, ensuring rapid service stabilization.
- Automated Kubernetes cluster setup for on-prem and cloud VMs using a Bash script wrapping Kubespray, Helm, and ingress controller installation, reducing cluster setup time by 45%.
- Automated environment setup across on-prem and cloud VMs using Ansible playbooks for service installation and configuration and Python and Bash scripts for environment automation tasks, cutting setup time by 50%.
- Configured SQL Server Always On Availability Groups for high availability and automatic failover across multiple production on-prem environments.
- Containerized and migrated legacy Ruby on Rails applications to AWS EKS with zero downtime; implemented GitOps-based CI/CD pipelines with Jenkins and Argo CD.
- Deployed and maintained observability stack (Prometheus, Grafana, Loki, Sentry); alerting rules reduced mean time to detection (MTTD) by 40%, improving incident response.
- Reduced CI/CD pipeline execution time by 40–50% through Docker layer caching, dependency caching, and build optimization in Azure DevOps and Jenkins.
- Reduced Azure infrastructure costs by 35% through environment consolidation, resource limits enforcement, VM and SQL Server right-sizing, and decommissioning idle resources.
- Conducted security audits across 10+ cloud and containerized environments, remediating over-privileged IAM roles, CVE-flagged container images, and critical misconfigurations across multiple projects.
- Performed blue/green production deployments to ensure zero-downtime releases and safe rollback for critical services for 3+ Ruby on Rails based projects.
- Delivered knowledge-sharing sessions on DevSecOps, Azure AI Services, and Linux for internal engineering teams.

DevOps Engineer Intern *08/2022 – 10/2022*

- Provisioned AWS infrastructure (VPC, EC2, S3) using Terraform, installed and configured Kubernetes and Jenkins across the environment.
- Built Jenkins CI/CD pipelines and containerized two applications, deploying them to Kubernetes in collaboration with developer interns.

Education

- **Arab Academy for Science, Technology, and Maritime Transport** Alexandria, Egypt
  Bachelor Degree in Computer Engineering *09/2017 – 02/2023*

Projects

- **Homelab**: Production-like Kubernetes homelab on Talos Linux with security enforced at every layer. Bootstrapped the full platform stack using Terraform, provisioning Cilium, Cert Manager, OpenEBS, Vault with Vault Secrets Operator, Authentik, and Argo CD with all configurations managed as IaC. Implemented GitOps using Argo CD with an App-of-Apps pattern, and deployed a full observability stack with kube-prometheus-stack, Loki, and Alloy, including custom alert rules on Kubernetes and Vault audit logs to detect critical security events.
- **Github Org Inviter**: CLI tool built with Elixir to automate GitHub organization access management, supporting user invites, removals, and team assignments via the GitHub API. Integrated Gemini AI to parse GitHub issue content into structured actions, enabling fully automated onboarding and offboarding workflows triggered through GitHub Actions.
- **Topics Extractor**: LangChain-based script deployed as a GCP Cloud Function, triggered automatically when a transcript file is uploaded to a Cloud Storage bucket, that extracts topics from video and audio transcripts using Gemini and generates relevant hashtags, with infrastructure managed via Terraform and secrets via GCP Secret Manager.
- **Survey Analyzer**: Demo web application built with Flask and React that extracts structured fields from PDF or image-based survey forms using a custom Azure Document Intelligence model, then applies sentiment analysis on feedback responses using Azure Language Services.